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标题	Atroposelective Ir-Catalyzed C – H Borylation of Phthalazine Heterobiaryls
作者	Stehrer P., Spannenberg A., Hapke M.
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摘要（原文节选）

The atroposelective iridium-catalyzed borylation of menthyloxy-substituted phthalazine heterobiaryls with diborons is reported. Utilizing [Ir(OMe)(COD)]2/2-aminopyridine as a rarely used efficient catalyst system, the heterobiaryls were selectively borylated in the 2-position of the carbocycle, exclusively yielding only one of the atropisomers, depending on the substitution of the phthalazine with (+)-menthyl or (−)-menthyl moieties. Exemplary further functionalization of a borylated atropisomer demonstrated that n …

数据表 · Table 1（9 行，含数值行 9 行）

Screening of Ligands for Directed Borylation Reaction with Substrate 3a

No.	Ligand	Yield [%]a
1	1,10-phenanthroline (L1)	(94)b
2	3,4,7,8-tetramethyl-1,10-phenanthroline (L2)	(67)b
3	2,2 ′ -bipyridine (L3)	(68)b
4	4,4 ′ -di-tert-butyl-2,2 ′ -bipyridine (L4)	(79)b
5	2-aminopyridine (L5)	88
6	2-aminomethylpyridine (L6)	n.r.c
7	2-(dimethyamino)-pyridine (L7)	74
8	2-(benzylamino)-pyridine (L8)	87
9	2-(benzylimino)-pyridine (L9)	68

脚注（逐字）：aIsolated yields.bReaction yielded mixture of indeterminable regioisomerically borylated heterobiaryls.cNo reaction and quantitative recovery of starting material.

正文对应句（逐字）：While (+)-menthyloxy substitution led to the (Ra)-configuration (compounds 4a, c, e, f, g, h), the introduction of a (−)-menthyloxy substituent led to the (Sa)-configuration (compounds 4b, d), which was derived from structure determination in the series of heterobiaryls 5 (vide infra). The isolated

数据表 · Table 2（8 行，含数值行 8 行）

Cross-coupling Reactions with Borylated Atropisomers

No.	Atropisomer/Aryl halide/Catalyst system/Conditions	Yield [%]	Stereoret. [% de]
1	4g, 4-Iodoacetophenone, Ni(COD)2, dppb, 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 12 h	75	25
2	4g, 4-Bromoacetophenone, Ni[P(n-Bu)3]2(COD), dppb, 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 12 h	81	74
3	4g, 4-Iodoacetophenone, Ni[P(n-Bu)3]2(COD), dppb, 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 12 h	82	75
4	4k-SI, 4-Iodoacetophenone, Ni[P(n-Bu)3]2(COD), dppb, 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 12 h	82	67
5	4g, 4-Iodoacetophenone, Ni[P(n-Bu)3]2(COD), 2-MeTHF/H2O, i-Pr2EtN, rt, 12 h	67	72
6	4g, 4-Iodoacetophenone, Ni(dppb)(COD), 2-MeTHF/H2O, i-Pr2EtN, rt, 12 h	85	71
7	4g, 4-Iodoacetophenone, Ni(dppb)(COD), 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 0.5 h	10	70
8	4g, 4-Iodoacetophenone, Ni(dppb)(COD), 2-MeTHF/H2O, i-Pr2EtN, 50 ° C, 2 h	33	69

正文对应句（逐字）：While (+)-menthyloxy substitution led to the (Ra)-configuration (compounds 4a, c, e, f, g, h), the introduction of a (−)-menthyloxy substituent led to the (Sa)-configuration (compounds 4b, d), which was derived from structure determination in the series of heterobiaryls 5 (vide infra). The isolated

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